FOMC Stance: Which AI System Should a Bond Desk Deploy?

Comparing a fine-tuned RoBERTa specialist with Gemini Flash-Lite and Gemini Flash

TASK

Hawkish · Dovish · Neutral

DATA

2,480 labelled FOMC sentences

EVALUATION

Quality · Safety · Speed · Cost

DS602 · Master of Data Science in Economics · SMU · 30 September 2026

1. Business Problem

The system converts central-bank language into a monetary-policy stance signal.

WHAT WE PREDICT

Each FOMC sentence is labelled:
- Hawkish → tighter policy / higher-rate bias
- Dovish → easier policy / lower-rate bias
- Neutral → descriptive / no clear direction

WHY IT MATTERS

A bond desk can aggregate sentence-level tags into a statement-level stance signal. A wrong-way call can affect positioning.

BUSINESS QUESTION

Which system gives a useful combination of classification quality, low wrong-way risk, low latency and manageable cost?

- Important asymmetry: hawkish ↔ dovish is treated as the most costly error.
- The project therefore evaluates more than accuracy: it adds business cost, latency, robustness and statistical uncertainty.
- The final decision is based on a pre-registered rule rather than choosing a model by eye.

What this slide does: establishes the real-world decision the model evaluation is trying to support.

2. Classification Task & Label Definitions

Every sentence is reduced to one of three mutually exclusive stance categories.

HAWKISH

Leans toward tighter policy or higher rates.
Examples: inflation concerns, stronger growth, possible tightening.

DOVISH

Leans toward easier policy or lower rates.
Examples: weak growth, weaker jobs, policy support.

NEUTRAL

Neither direction, or purely descriptive.
Examples: historical facts, procedures, general economic description.

Example flow

INPUT

“Inflation remains elevated and further tightening may be appropriate.”

→

MODEL

RoBERTa / Gemini

→

OUTPUT

HAWKISH

What this slide does: makes the NLP task and the meaning of the three labels clear before discussing the models.

3. AI Systems Compared

The central experiment compares a domain-specialist model with two zero-shot LLMs.

ID	SYSTEM	APPROACH	INFERENCE	ROLE
S1	RoBERTa-base	Fine-tuned on 1,984 FOMC train sentences	Colab T4 GPU	Specialist
S2	Gemini 3.5 Flash-Lite	Zero-shot; shared prompt	API	General LLM
S3	Gemini 3.5 Flash	Zero-shot; shared prompt	API	General LLM
B1	Majority	Always predict neutral	—	Naive baseline
B2	Random	Uniform random label	—	Chance baseline

S1

4 epochs · LR 2e-5 · batch 16 · max length 128

S2 / S3

Same prompt, temperature 0, one-word answer

FAIR COMPARISON

All main systems score the same 496 held-out sentences

What this slide does: explains exactly which AI systems are being compared and why each one is included.

4. Dataset & Evaluation Pipeline

The notebook uses one fixed test set and runs every system through the same measurement pipeline.

2,480 TOTAL

1,984 train
496 held-out test

CLASS BALANCE

Neutral 50%
Dovish 27% · Hawkish 23%

CONTROL

Seed 42
Stratified sampling

REPRODUCIBILITY

Drive caching
0% parse failures

- 1. Load the Hugging Face FOMC benchmark.
- 2. Train S1 on the training split only; call S2 and S3 with the shared prompt.
- 3. Run all systems on the same 496 test sentences and cache every response.
- 4. Compute quality, business cost, latency, confidence and statistical tests.
- 5. Run stress tests: short/long, adversarial, multilingual, ECB OOD and fresh 2026 text.

What this slide does: shows that the comparison is reproducible and that no test sentence is privileged to a particular model.

5. Evaluation Metrics

Each metric answers a different question about whether a system is deployable.

QUALITY

Macro F1, accuracy, precision, recall, per-class F1 and confusion matrix.

SAFETY

Direction-flip rate and business cost. Flip = hawkish↔dovish.

EFFICIENCY

p50 / p95 / p99 latency, throughput, tokens and USD cost.

STATISTICS

2,000 paired bootstrap resamples, McNemar test and Holm correction.

ROBUSTNESS

Typos, hedges, negation, language changes, OOD and fresh text.

CALIBRATION

ECE, AUROC and AUPRC test whether confidence is useful for routing.

Why Macro F1? Neutral is 50% of the test set. Always predicting neutral gives about 0.50 accuracy but only 0.22 Macro F1, so Macro F1 is the headline quality metric.

What this slide does: connects each evaluation metric to a concrete deployment question.

6. How the Models Are Run

S1 learns from labelled FOMC data; S2/S3 classify by instruction prompting.

S1 · FINE-TUNED RoBERTa

roberta-base → add 3-class classification head → train on 1,984 sentences → validate on 10% of the training data → save the fitted model.

Training: 4 epochs, AdamW, LR 2e-5, batch 16, max length 128.

S2 / S3 · ZERO-SHOT GEMINI

The same prompt defines hawkish, dovish and neutral and asks for exactly one word.

Temperature = 0. Responses are parsed; invalid answers count as failures. Parse/API failures were 0%.

S1 architecture

FOMC sentence → tokenizer → RoBERTa transformer → 3-class head → probabilities → predicted label

Gemini prompting

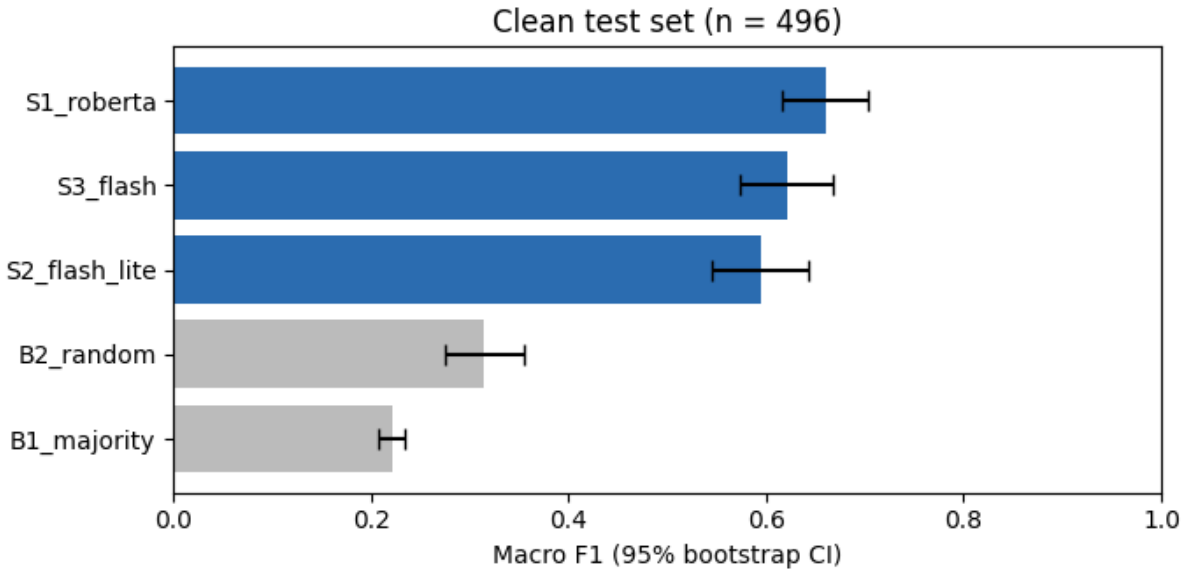
FOMC sentence + label definitions → Gemini → one-word label → deterministic parser

Important caveat: S2 uses default thinking because Flash-Lite rejects a zero thinking budget; S3 uses thinking off. This is a fairness limitation in the original notebook.

What this slide does: explains the model-building and inference process you may be asked about during Q&A.

7. Overall Quality: Macro F1

This is the main quality comparison on the clean 496-sentence test set.



SYSTEM	MACRO F1	95% CI	ACCURACY	FLIP RATE
S1 · RoBERTa	0.660	0.616–0.703	0.679	6.2%
S2 · Flash-Lite	0.595	0.545–0.643	0.645	2.8%
S3 · Flash	0.622	0.573–0.668	0.651	2.8%
S1→S3 cascade	0.676	—	—	2.8%

READ THE CHART

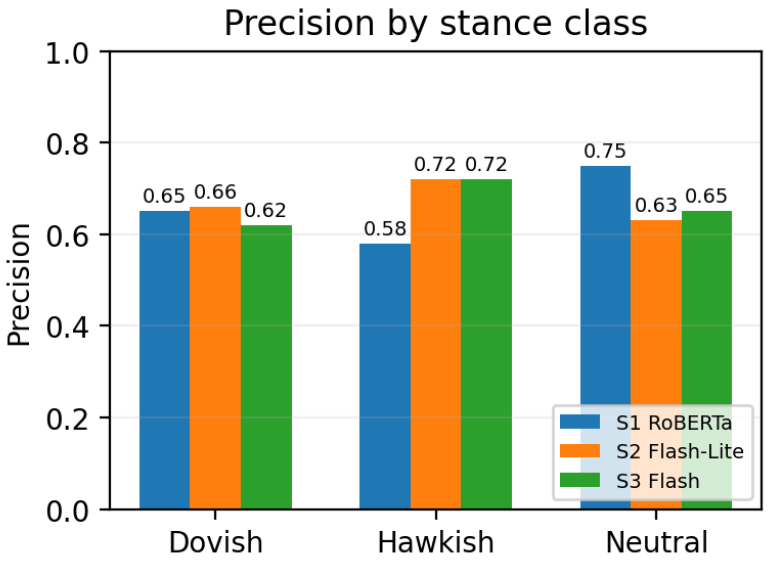
S1 has the highest single-model Macro F1. The cascade has the highest observed F1, but it is slower.

Chart: Macro F1 with 95% bootstrap confidence intervals from the Colab.

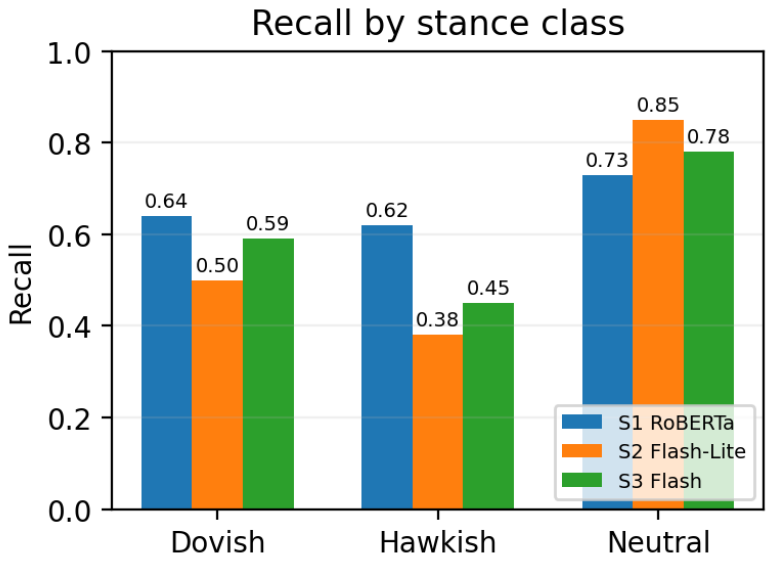
What this slide does: gives the headline quality result before we break it down into precision, recall and individual classes.

8. Precision, Recall & F1

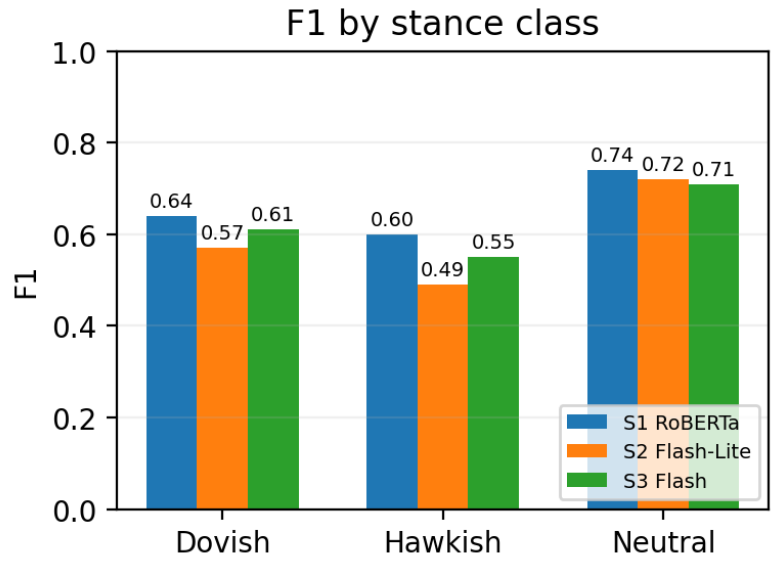
These three charts explain where each model gains or loses performance by stance class.



Precision: when the model predicts a class, how often is that prediction correct?



Recall: of all true examples of a class, how many does the model detect?



F1: harmonic balance between precision and recall for each class.

MAIN PATTERN

Gemini's neutral recall is high (0.85 / 0.78), but hawkish recall is low (0.38 / 0.45).

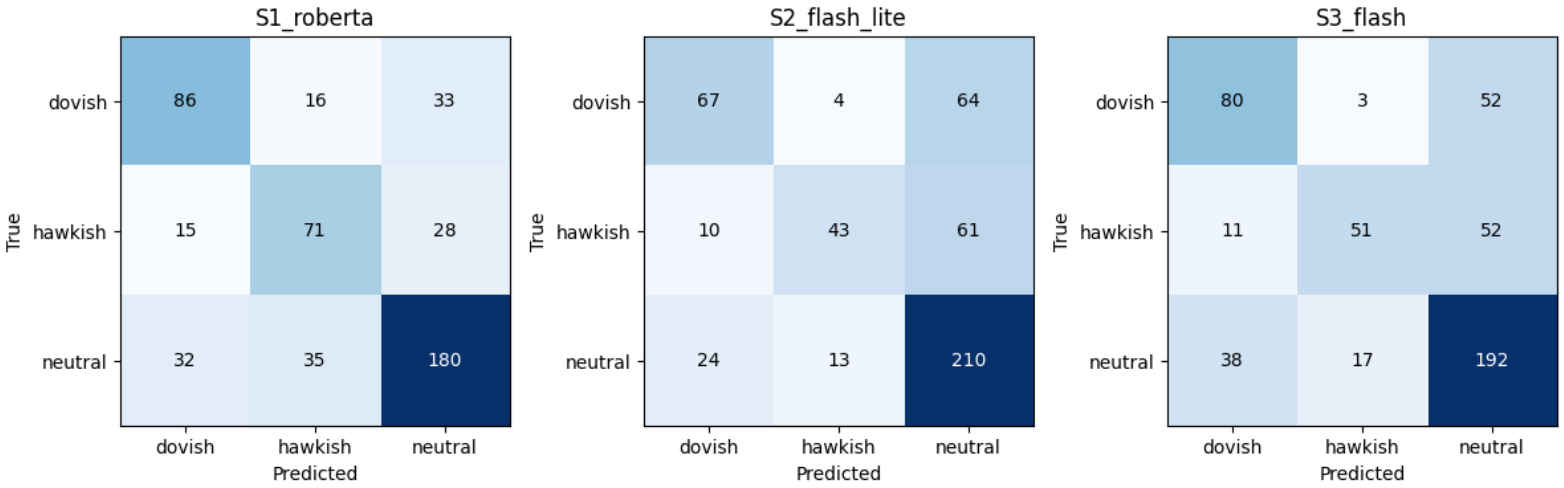
S1 PATTERN

S1 is more balanced: recall ranges from 0.62 to 0.73 across the three classes.

What this slide does: shows why Macro F1 differs — Gemini often chooses neutral, while S1 commits to stance labels more often.

9. Confusion Matrices

The confusion matrix shows exactly which labels are being confused.



S1 · RoBERTa

Correct cells dominate the diagonal. Some dovish/hawkish swaps explain its 6.2% flip rate.

S2 · Flash-Lite

A large share of hawkish examples are predicted as neutral — the “cautious” failure mode.

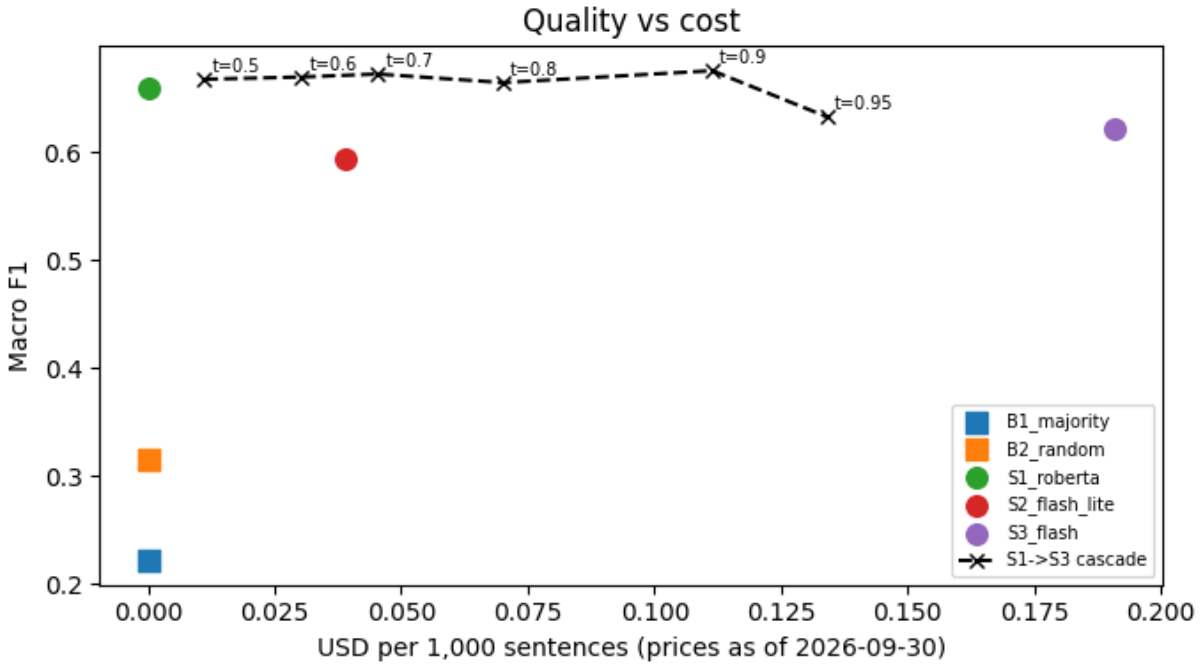
S3 · Flash

Similar pattern: strong neutral recall but a substantial number of hawkish cases are called neutral.

What this slide does: turns the overall scores into specific error patterns that are easier to interpret than one number.

10. Cost, Latency & Throughput

The desk also cares about whether a good model is affordable and fast enough to use live.



SYSTEM	P50	P95	P99	SENT./S	USD/1K
S1	0.009s	0.012s	0.016s	105.9	0*
S2	0.762s	0.961s	1.152s	1.32	0.039
S3	1.283s	1.975s	2.839s	0.72	0.191
Cascade	—	1.741s	—	—	0.111

HOW TO READ THE PLOT

X-axis = USD per 1,000; Y-axis = Macro F1. The dashed cascade shows different confidence thresholds.

KEY TRADE-OFF

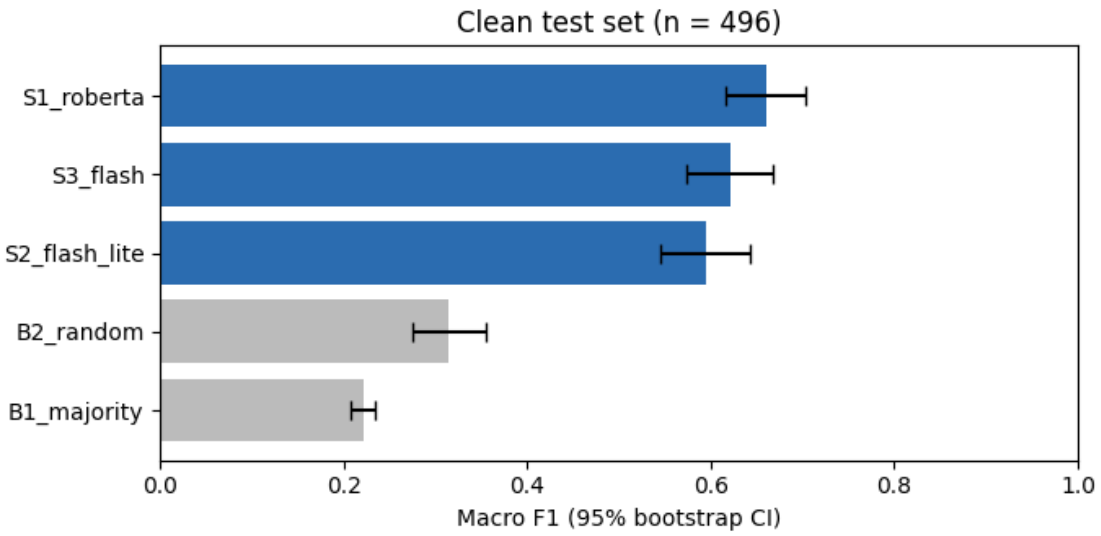
S1 is dramatically faster. Gemini costs more but produces fewer costly flips. S1's GPU cost is excluded from the USD column.

*S1 has no API bill but requires GPU time. S2's displayed API cost excludes thinking tokens.

What this slide does: connects model quality to the practical resources needed to run the system.

11. Statistical Significance

The notebook tests whether observed differences are large enough to be distinguished from sampling variability.



PAIRED BOOTSTRAP

The same 496 sentences are resampled 2,000 times for both systems. The middle 95% of the F1 gap gives a paired confidence interval.

MCNEMAR'S TEST

Looks only at sentences where two models disagree on correct/incorrect status. It tests whether one model wins those disagreements systematically.

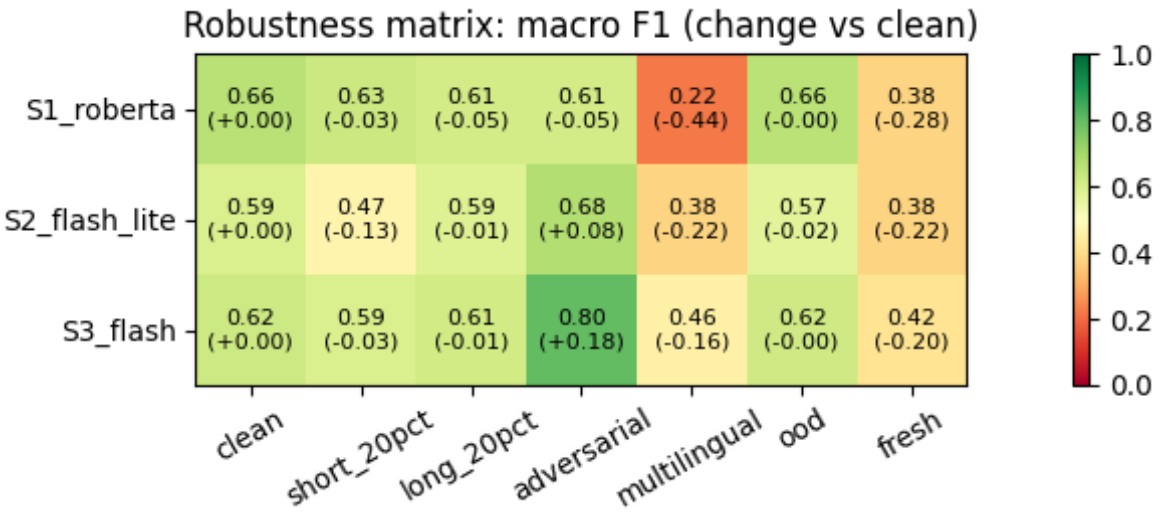
HOLM CORRECTION

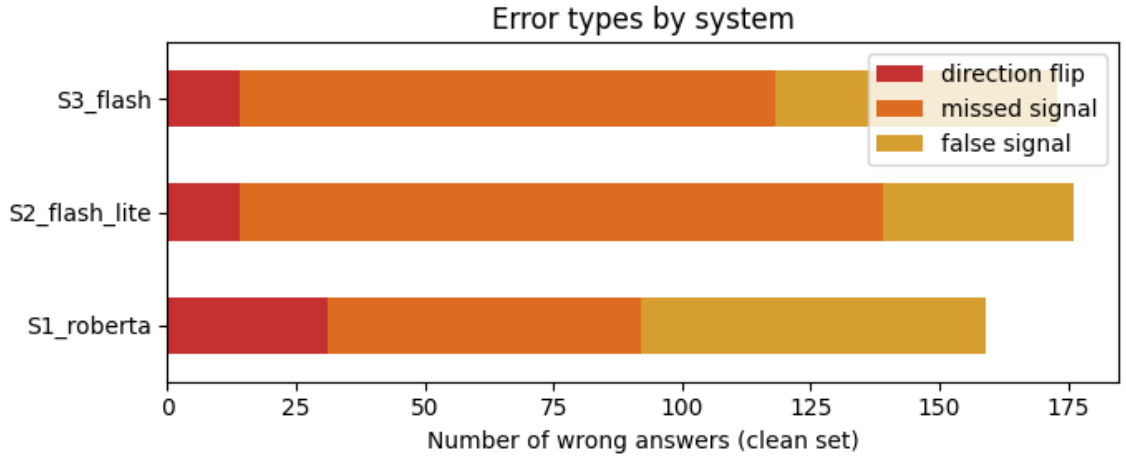
There are three pairwise comparisons. Holm adjusts the p-values to control the false-positive risk from multiple tests.

Result: S1 vs S2 and S1 vs S3 show positive observed F1 gaps, but no pair remains statistically significant after Holm correction.

12. Robustness & Error Analysis

The project tests whether clean-test performance survives harder language and distribution shifts.





MULTILINGUAL

S1 drops to 0.22 Macro F1 on Hindi/Chinese; Gemini performs better on this set.

ERROR TYPES

S1 makes fewer total errors but more direction flips. Gemini's errors are dominated by missed signals.

LABEL QUALITY

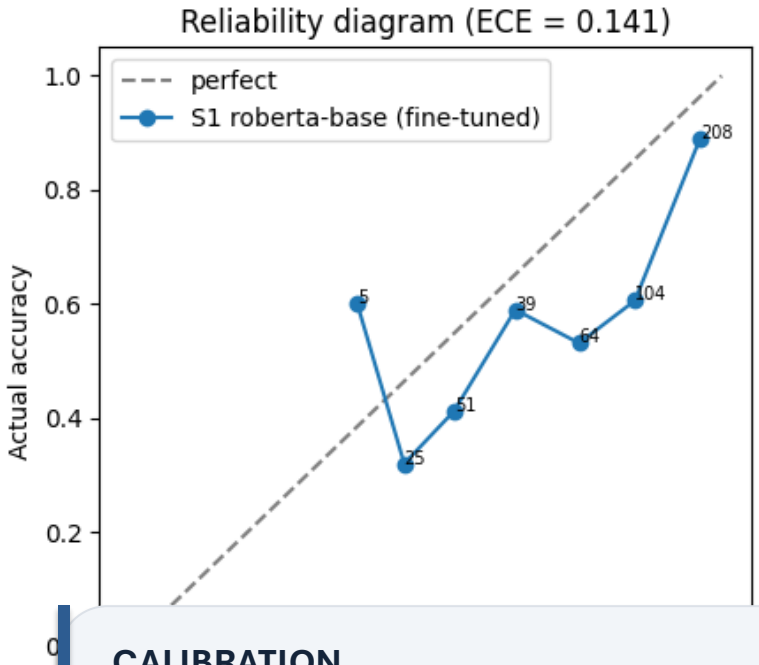
Blind relabelling: 58% agreement with gold; Cohen's κ = 0.31. Some gold labels are disputable.

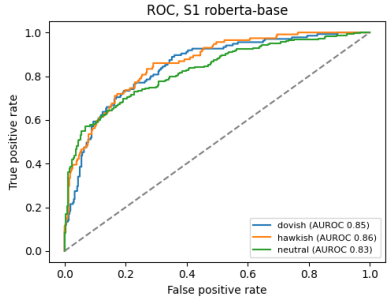
Heatmap: Macro F1 under robustness tests. Bar chart: wrong-answer types on the clean test set.

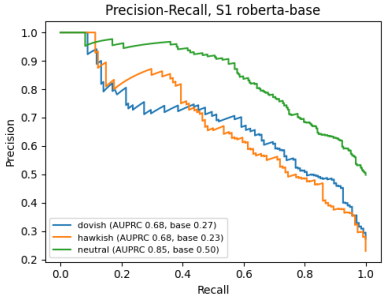
What this slide does: checks generalization and explains what kinds of mistakes actually occur.

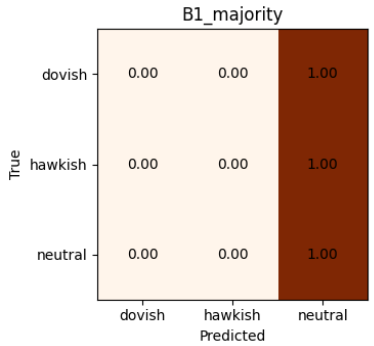
13. Calibration, ROC/PR & Real-World Check

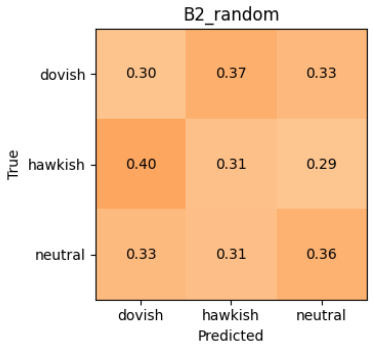
These analyses go beyond basic classification accuracy.











CALIBRATION

ECE = 0.141. S1 is somewhat overconfident, so confidence must be checked before using it for cascade routing.

ROC / PRECISION-RECALL

AUROC/AUPRC measure discrimination across thresholds. For S1: dovish AUROC ≈ 0.85, hawkish ≈ 0.86, neutral ≈ 0.83.

STANCE INDEX

Sentence tags are aggregated as (hawkish − dovish) / total. It is a sanity check against actual FOMC decisions, not a forecast.

What this slide does: shows why confidence quality, class discrimination and statement-level behaviour matter for a deployment system.

PRE-REGISTERED RULE

Candidate must be within 2 Macro-F1 points of the best, have p95 < 1 second, and have a flip rate not significantly worse than the best.

RULE OUTCOME

No candidate meets all three conditions. The fallback therefore selects the highest observed Macro F1: the S1 → S3 cascade at t = 0.9.

S1 ALONE

Fastest: p95 12 ms
Best single-model Macro F1: 0.660
But 6.2% direction flips

GEMINI

Fewer flips: 2.8%
Lower desk cost than S1
S3 is too slow for the 1s target

CASCADE

Highest observed F1: 0.676
Lowest desk cost: 415 / 1k
But p95 = 1.741s

- Before production: tune the cascade threshold on validation data, solve the latency constraint, match Gemini thinking settings, and expand fresh labelled data.
- Limitations: small stress sets, one annotator, test-set threshold tuning, and Colab network latency.
- Statistical takeaway: observed differences are not significant after Holm correction.

FINAL MESSAGE: The project is not asking "Which model has the highest score?" It asks which system provides an acceptable balance of quality, wrong-way risk, speed and cost.